Task 5: Set Up Version Control Using Git

Objective

To implement version control using Git by creating a repository, committing changes, creating branches, merging branches, and pushing the project to GitHub.

```
admin1@DESKTOP-BBG39NK:~$ sudo apt install git -y
git is already the newest version (1:2.53.0-1ubuntu1).
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 159
admin1@DESKTOP-BBG39NK:~$ git version
git version 2.53.0
```

Repository Creation

A Git repository was initialized using: `git init`

```
admin1@DESKTOP-BBG39NK:~/devops-ci-cd-lab$ git init
hint: Using 'master' as the name for the initial branch. This default branch name
hint: will change to "main" in Git 3.0. To configure the initial branch name
hint: to use in all of your new repositories, which will suppress this warning,
hint: call:
hint:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint:   git branch -m <name>
hint:
hint: Disable this message with "git config set advice.defaultBranchName false"
Initialized empty Git repository in /home/admin1/devops-ci-cd-lab/.git/
admin1@DESKTOP-BBG39NK:~/devops-ci-cd-lab$ git status
On branch master

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        app.py

nothing added to commit but untracked files present (use "git add" to track)
```

The project files were then staged and committed:

```
admin1@DESKTOP-BBG39NK:~/devops-ci-cd-lab$ git add .
admin1@DESKTOP-BBG39NK:~/devops-ci-cd-lab$ git commit -m " DevOps application"
[master (root-commit) 7be3151]  DevOps application
 1 file changed, 1 insertion(+)
 create mode 100644 app.py
```

Branching

A separate development branch was created: `git switch -c development`

```
admin1@DESKTOP-BBG39NK:~/devops-ci-cd-lab$ git branch development
admin1@DESKTOP-BBG39NK:~/devops-ci-cd-lab$ git switch development
Switched to branch 'development'
admin1@DESKTOP-BBG39NK:~/devops-ci-cd-lab$ git add .
admin1@DESKTOP-BBG39NK:~/devops-ci-cd-lab$ git commit -m "Update application greeting"
On branch development
```

A change was made and committed on the development branch:

```
git add README.md
git commit -m "Update README on development branch"
```

Merging

The main branch was selected: `git switch main`

The development branch was merged into main: `git merge development`

```
admin1@DESKTOP-BBG39NK:~/devops-ci-cd-lab$ git merge development
Already up to date.
admin1@DESKTOP-BBG39NK:~/devops-ci-cd-lab$ git log --oneline --graph --all
* 7be3151 (HEAD -> development, master)  DevOps application
admin1@DESKTOP-BBG39NK:~/devops-ci-cd-lab$ git remote add origin https://github.com/gurleen-51/devops-ci-cd-lab.git
admin1@DESKTOP-BBG39NK:~/devops-ci-cd-lab$ git remote -v
origin  https://github.com/gurleen-51/devops-ci-cd-lab.git (fetch)
origin  https://github.com/gurleen-51/devops-ci-cd-lab.git (push)
```

Commit History

The Git history was inspected using: `git log --oneline --graph --all`

Finally, the updated main branch was pushed to GitHub: `git push origin main`

```
admin1@DESKTOP-BBG39NK:~/devops-ci-cd-lab$ git commit -m "Initial commit"
On branch development
nothing to commit, working tree clean
admin1@DESKTOP-BBG39NK:~/devops-ci-cd-lab$ git branch -M main
admin1@DESKTOP-BBG39NK:~/devops-ci-cd-lab$ git branch
* main
  master
admin1@DESKTOP-BBG39NK:~/devops-ci-cd-lab$ git push -u origin main
```

The project was hosted in the GitHub repository.

Git successfully provided version control for the project. Branching and merging allowed development changes to be managed separately before integrating them into the main branch.